

Junior Makers · Class 15

Pong Game with a Knob Controller

Build the game AND the controller!

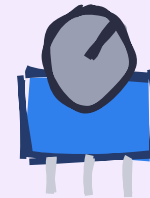
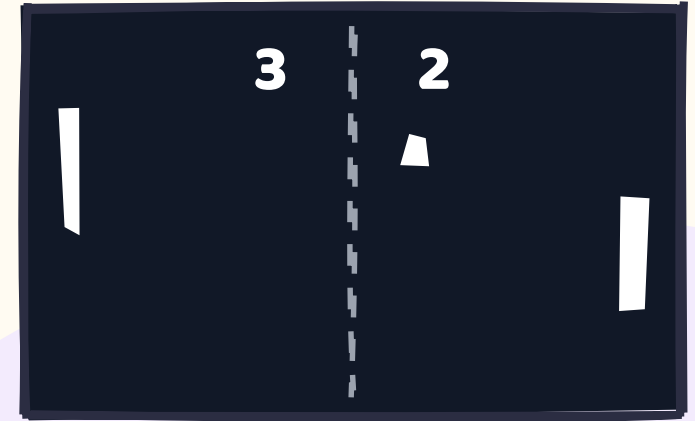
Potentiometer

Mapping numbers

Game loop

Did you know?

Pong came out in 1972. The first machine stopped working after two days because it was too FULL of coins!





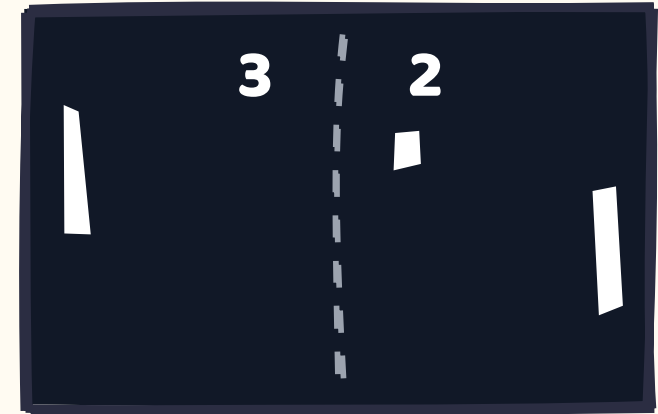
Let's think...

How can a knob play a video game?



1972: one knob, one ball

In the first Pong machines, players turned a simple **knob** to move a paddle up and down. No buttons, no joystick, no touch screen. How does twisting a knob become **movement on a screen**?



Magic wire?

Tiny motor?

A number!



Today: we build Pong in **mBlock** and make our own **knob controller** with an Arduino.





Learning objectives

By the end of this class I can...



-  **Knob to number:** explain how a **potentiometer** turns a twist into a number from 0 to 1023 **Proof:** my knob readings at both ends and the middle
-  **Map:** **map** the knob number to the paddle's position on the screen **Proof:** turning the knob moves my paddle from top to bottom
-  **Game loop:** make the ball move and **bounce** using a forever loop and if rules **Proof:** my ball bounces off the walls and my paddle
-  **Playtest:** **playtest** my game with a friend and improve it **Proof:** one change I made after my friend's feedback





Objective 1 · Knob to number · video

Video time!



The Evolution of Video Games: Pong's 50-Year Legacy

The Sydney Morning Herald · about 7 minutes · youtu.be/763CFRuxovo

Watch for...

- 1 What **year** did Pong come out, and where did people play it?
- 2 How was Pong **controlled**?
- 3 Name **one way** games have changed since Pong.



After the video: With your partner, list **3 rules** of Pong. A computer has to follow every one of those rules!





Objective 1 · Knob to number

A potentiometer turns a twist into a number



The idea

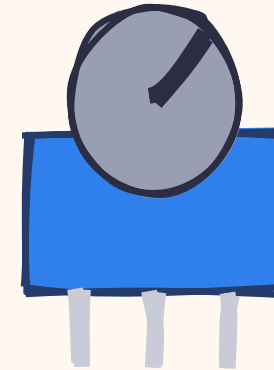
Inside the knob is a strip of carbon and a little slider. Turning the knob moves the slider, so the **voltage** on the middle pin changes from **0 V** to **5 V**.

The Arduino reads that voltage on pin **A0** as a number from **0** to **1023**. Turn it all the way left: 0. All the way right: 1023.

10k?

10k is the knob's resistance: 10,000 ohms. That is printed on the back.

Example: a volume knob



0
↓
1023

Turning the volume knob on a speaker or car radio does the same thing. A little twist = a little louder. The knob does not jump between 'on' and 'off': it can be **anywhere** in between. That is called **analog**.



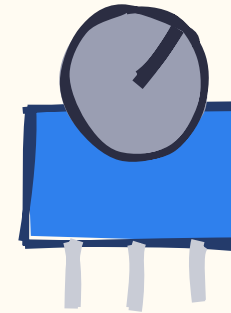


Objective 1 · Knob to number · test it

Read your knob



Knob position	Our reading	Prediction
All the way left		0?
Half way		
All the way right		1023?
Where you think 250 is		

**knob (10k)****Smooth?**

Does the number go up smoothly, or jump? Turn slowly and watch.

**forever**

set knob to read analog pin A0

say knob

Live mode

In mBlock choose the **Arduino Uno** device and switch to **Live**. The sprite can then read the knob straight away!



Wiring: outer legs to **5V** and **GND**, middle leg to **A0**. Backwards? Swap the outer legs!



Real world

Knobs and sliders all around us



Joysticks

Two hidden knobs inside: one for left-right, one for up-down



Car steering

Sensors tell the car how far the wheel is turned



Mixing desks

Sliders set how loud each instrument is



Oven dials

Turn to choose the temperature



Find a knob or slider in your classroom. What number could it be controlling?

Analog vs digital

A button is digital: only 0 or 1. A knob is analog: any value in between.



Class game

Human Pong



How to play

1. Two players sit at the ends of a desk, each holding a notebook as a paddle.
2. Gently bat a scrunched-up paper ball across the desk.
3. The class are the **computer**: shout the rule every time something happens.
4. **"Wall: bounce!" · "Paddle: bounce back!" · "Missed: point!"**
5. First to 5 points wins. Swap players.

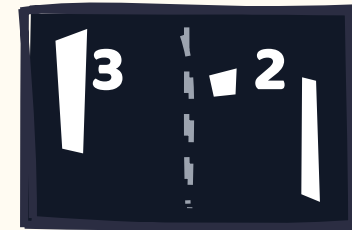
Then discuss

Which rules did the 'computer' need to know? Write them as **if ... then ...** sentences.



Big idea

A video game is just a list of simple rules checked again and again, very fast.





Build it

Our Pong kit (per group)



Arduino Uno

The controller's brain



Potentiometer

The paddle knob



Push button

Serve the ball (+ 10k)



LED + 220 Ω

Lights up on a point



Buzzer

Beeps on a bounce



Laptop

mBlock 5 + the game

Also: breadboard and jumper wires. Put the knob and button in a small box or cup: now it is a real controller!

Scratch first

No Arduino? Build the game in Scratch with arrow keys first, then add the knob.



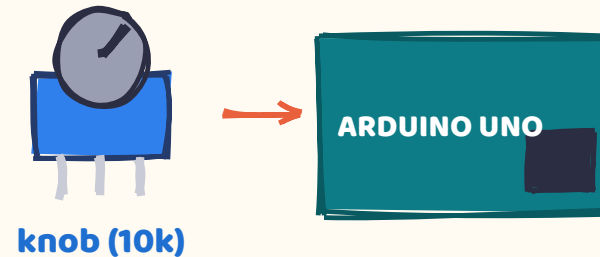


Objective 1 · Knob to number · Build step 1 of 4

Wire the controller



Part	Connect to
Knob outer legs	5V and GND
Knob middle leg	A0
Button	pin 2 and 5V
10k resistor	pin 2 to GND (pull-down)
LED + 220 Ω	pin 12 to GND
Buzzer +	pin 11



Why the 10k?

Without it, pin 2 'floats' and reads random 0s and 1s. The 10k gently holds it at 0 until you press.

Controller box

Poke the knob through a paper cup. Hold it like a real game controller!





Objective 2 · Map

Mapping: change one range into another



The idea

The knob gives **0 to 1023**. The paddle must go from **y = -150** (bottom) to **y = 150** (top) on the stage. We need to **map** one range onto the other:

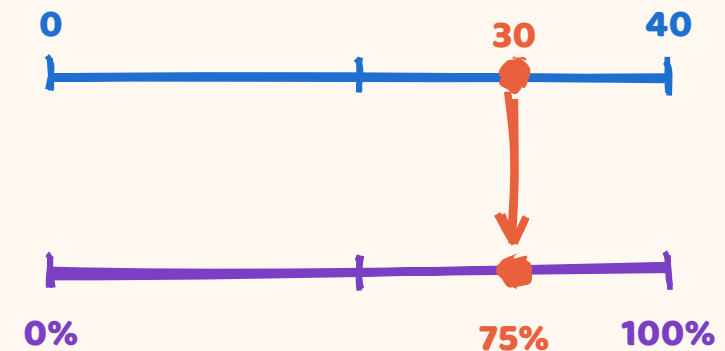
$$\text{paddle } y = \text{knob} \div 1023 \times 300 - 150$$

Knob 0 → -150 (bottom). Knob 1023 → 150 (top). Knob 512 → about 0 (middle).

Shortcut

mBlock and Arduino also have a ready-made map block. Try it after you understand the maths!

Example: test marks to percent



You scored **30 out of 40** in a test. To get a percentage: $30 \div 40 \times 100 = 75\%$. You mapped the range 0 to 40 onto the range 0 to 100. Same maths, different numbers!



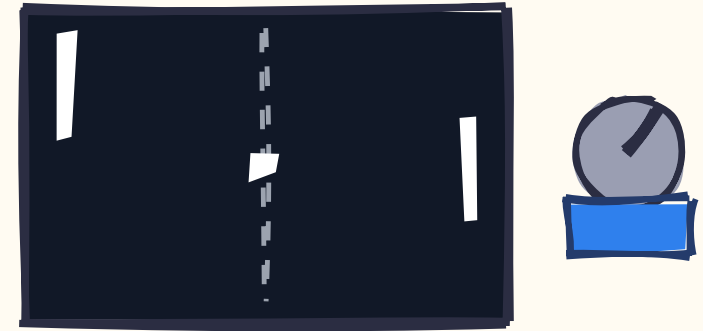


Objective 2 · Build step 2 of 4

Make the paddle follow the knob



- a** Draw a **Paddle** sprite: a thin rectangle, about 10×70 pixels.
- b** Set its x to **-220** (left side of the stage).
- c** Add: **forever** → **set y to $(\text{knob} \div 1023 \times 300 - 150)$** .
- d** Turn the knob slowly. Does the paddle reach the top AND the bottom?
- e** Paddle moves the wrong way? Change the formula to **$150 - \text{knob} \div 1023 \times 300$** .



Check: Turning the knob moves the paddle smoothly from top to bottom.



Use **Live** mode, not Upload: the sprites live on the laptop.





Objective 3 · Game loop

A game loop: move a little, check, repeat



The idea

Every game runs a **game loop** many times a second:

1. **Move** the ball a few steps.
2. **Check** the rules: hit a wall? hit the paddle? missed?
3. **Draw** the new picture.

Then do it all again. Fast loops make smooth movement.

FPS

Most games loop 30 to 60 times every second. That is called frames per second (FPS).

Example: a flipbook



tiny move each page

Draw a ball on each page of a notebook, moved a tiny bit each time. Flick the pages fast and the ball **moves**! Each page is one turn of the game loop.





Objective 3 · Game loop

Bouncing is just an if rule



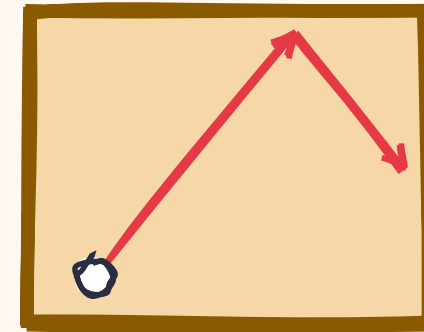
The idea

if the ball touches the **top or bottom edge** → bounce (Scratch: **if on edge, bounce**).

if the ball touches the **paddle** → turn round: **point in direction (0 - direction)**.

if the ball goes past the paddle → the other side scores a point and the ball goes back to the middle.

Example: a carrom striker



same angle in, same angle out

Flick the striker against the side of the carrom board. It bounces off at the **same angle** it came in. Our ball follows that rule too.



Objective 3 · Game loop · Build step 3 of 4

Code the ball

 when green flag clicked

go to x: 0 y: 0 · point in direction (pick random 45 to 135)

forever

move 8 steps

if on edge, bounce

if touching Paddle → point in direction (0 - direction) · move 10 steps

if x position < -230 → change computer score by 1 · go to x: 0 y: 0



The other side



Draw a wall or a computer paddle on the right that always bounces the ball back.
Then you play alone against the computer!

Stuck to the paddle?

Add move 10 steps after the bounce so the ball escapes before the next check.



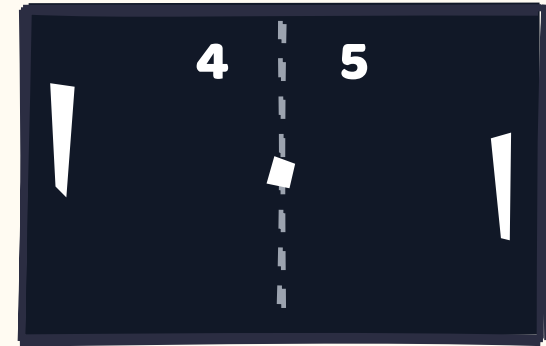


Objective 3 · Build step 4 of 4

Serve, score and sound



- a** Make two variables: **my score** and **computer score**. Tick them to show on the stage.
- b** Ball waits in the middle until the **button** on pin 2 is pressed. Then it serves!
- c** On a bounce off your paddle: **beep** the buzzer on pin 11.
- d** When you score: light the **LED** on pin 12 for 1 second.
- e** First to **5** wins: show a 'You win!' or 'Try again' message.



Check: The button serves, the score changes and the game ends at 5.





Objective 4 · Playtest

Playtest like a game designer



The idea

Game designers watch **real players** try their game. They do not explain anything: they watch where players get stuck or bored, then improve the game.

Swap seats with another team. Play their game for **3 minutes**, then fill in their feedback card.

Feedback card	Answer
Fun? (1 to 5)	
Too easy or too hard?	
One thing I loved	
One thing to change	
Our change after feedback	



Example: testers said "the ball is too slow and boring". The team made the ball speed up by 1 step after every bounce. Now every rally gets more exciting!





Level up!

Game designer challenges



Choose your challenge



Two players: add a second knob on A1 for a friend's paddle on the right.



Speed up: the ball gets 1 step faster after every bounce.



Power-up: a star appears; hit it to make your paddle bigger for 10 seconds.



Sound design: different beeps for walls, paddles and points.

Think like an engineer

Why do you think a knob was better than buttons for the first Pong machines?



Boss level

Make the bounce angle depend on WHERE the ball hits the paddle, like real Pong.





Objective check

Can I really do it? Prove it.



01 Turn your knob and read out 3 numbers. What are the smallest and biggest possible?



02 Explain the paddle formula. What y does the paddle get when the knob reads 1023?



03 Point to the **if** rule that makes the ball bounce off your paddle.



04 Tell us one change you made after another team playtested your game.



Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Hold a mini tournament: winners of each table play each other on the projector.





Quick quiz

Fill the gaps



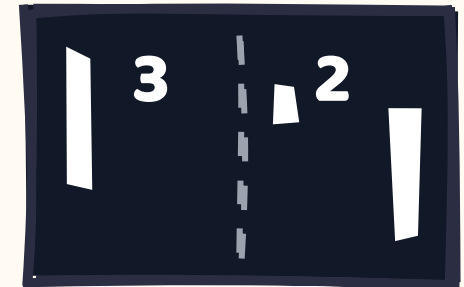
Question

Answer

1. The Arduino reads the knob as a number from 0 to ____.
2. A knob is analog; a button is ____.
3. Knob 0 → paddle y = ____ (top or bottom?)
4. Move, check, draw, repeat is called the game ____.
5. Testing a game with real players is called ____.

Answers

Cover this! 1: 1023 · 2: digital · 3: -150, bottom · 4: loop · 5: playtesting



Show what you know

What did we learn?

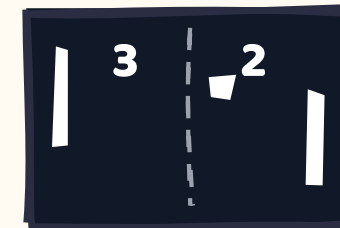
- ✓ A potentiometer turns a twist into 0 to 1023
- ✓ Mapping changes one range into another
- ✓ A game loop moves, checks rules and repeats
- ✓ Bouncing is just an if rule
- ✓ Playtesting makes games better



Exit ticket: the knob reads 0. Where is the paddle, and why?

At home

Teach someone at home how to play your Pong game. What did they find hard?



Next class: build an AI Recycling Sorter! ✨